

UNIT III – THEORY OF LOGICS & PREDICATE LOGIC

III Theory of Logics: Proposition, Truth tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Theory of Inference. Predicate Logic: First order predicate, well-formed formula of predicate, quantifiers, Inference theory of predicate logic.

◇ 1. Theory of Logics

Logic helps us **analyze and reason** about statements and arguments.

In digital logic, it is used to represent how **circuits and programs make decisions** (True/False).

🔗 1.1 Proposition

Definition:

A **proposition** is a **declarative statement** that is either **True (T)** or **False (F)** — not both.

✅ Examples:

- “ $2 + 2 = 4$ ” $\rightarrow$ True
- “Delhi is in India” $\rightarrow$ True
- “ $5 < 3$ ” $\rightarrow$ False

❌ Not propositions:

- “What is your name?” (a question)
- “Go home!” (a command)

◇ 1.2 Logical Connectives

Logical connectives combine propositions to form **compound statements**.

Sym bol	Name	Expre ssion	Read As	Output True When
$\neg$	Negation (NOT)	$\neg P$	not P	P is False
$\wedge$	Conjunction (AND)	$P \wedge Q$	P and Q	both P, Q are True
$\vee$	Disjunction (OR)	$P \vee Q$	P or Q	at least one is True
$\rightarrow$	Implication	$P \rightarrow Q$	if P then Q	False only if P=True, Q=False
$\leftrightarrow$	Biconditional	$P \leftrightarrow Q$	P if and only if Q	both same (T,T or F,F)

1.3 Truth Tables

Example 1:

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

1.4 Tautology, Contradiction, and Satisfiability

Type	Definition	Example	Result
Tautology	A statement always True for all truth values of variables.	$(P \vee \neg P)$	Always True
Contradiction	Always False , no matter what the values are.	$(P \wedge \neg P)$	Always False
Satisfiable	Sometimes True, sometimes False depending on values.	$(P \wedge Q)$	Depends on P, Q

Use in Circuits:

- Tautology $\rightarrow$ output always 1 (ON)
- Contradiction $\rightarrow$ output always 0 (OFF)

◇ 1.5 Algebra of Propositions (Logical Laws)

These are similar to Boolean algebra laws.

Law	Formula	Description
Identity	$P \wedge T = P, P \vee F = P$	No change with identity values
Null	$P \wedge F = F, P \vee T = T$	Dominant values
Idempotent	$P \wedge P = P, P \vee P = P$	Repetition doesn't change result
Complement	$P \wedge \neg P = F, P \vee \neg P = T$	Opposite cancels
Commutative	$P \vee Q = Q \vee P, P \wedge Q = Q \wedge P$	Order doesn't matter
Associative	$(P \vee Q) \vee R = P \vee (Q \vee R)$	Grouping doesn't matter
Distributive	$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$	Distribution property
De Morgan's	$\neg(P \wedge Q) = \neg P \vee \neg Q, \neg(P \vee Q) = \neg P \wedge \neg Q$	Conversion between AND $\leftrightarrow$ OR

1.6 Theory of Inference (Logical Reasoning)

Inference means **deriving new statements** (conclusions) from known facts (premises).

Example:

1. If it rains, the ground gets wet. $\rightarrow (P \rightarrow Q)$
2. It rains. $\rightarrow (P)$

$\therefore$ Therefore, the ground gets wet. $\rightarrow (Q)$

This is **Modus Ponens** (method of affirmation).

Common Rules of Inference

Rule	Form	Explanation
Modus Ponens	$(P \rightarrow Q, P \Rightarrow Q)$	If P then Q
Modus Tollens	$(P \rightarrow Q, \neg Q \Rightarrow \neg P)$	If not Q, then not P
Hypothetical Syllogism	$(P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R)$	Chain implication
Disjunctive Syllogism	$(P \vee Q, \neg P \Rightarrow Q)$	If one false, other true
Addition	$(P \Rightarrow P \vee Q)$	Adding an OR condition
Simplification	$(P \wedge Q \Rightarrow P)$	From AND, one part true
Conjunction	$(P, Q \Rightarrow P \wedge Q)$	Combine two truths

◇ 2. Predicate Logic

Definition

Predicate logic extends propositional logic by including **variables and quantifiers**. It allows reasoning about **objects and their properties**.

2.1 Predicate and Proposition

Basis	Propositional Logic	Predicate Logic
Components	Simple True/False statements	Statements with variables
Example	"It is raining."	"x is greater than 5."
Variable	None	Contains variable (x)
Use	Simple logic	Advanced logic reasoning

2.2 Predicate

A **predicate** is a **statement involving variable(s)** that becomes a proposition when specific values are given.

Example:

Let $(P(x))$: "x is even."

- If $x = 2 \rightarrow \text{True}$

- If $x = 3 \rightarrow \text{False}$

So, $(P(x))$ depends on the value of x .

◇ 2.3 Quantifiers

Quantifiers tell us **how many elements** of a domain satisfy the predicate.

Symbol	Name	Meaning	Example
$\forall$	Universal Quantifier	“For all”	$(\forall x, P(x))$: all x satisfy P
$\exists$	Existential Quantifier	“There exists”	$(\exists x, P(x))$: at least one x satisfies P

💡 **Example:**

- $(\forall x (x^2 \geq 0)) \rightarrow \text{True for all real numbers}$
- $(\exists x (x^2 = 4)) \rightarrow \text{True } (x = 2 \text{ or } -2)$

📖 2.4 Well-Formed Formula (WFF)

A **Well-Formed Formula** is a **logically correct expression** that follows the rules of syntax in predicate logic.

Rules:

1. Every proposition and predicate is a WFF.
2. If P is a WFF, then $\neg P$ is also WFF.
3. If P and Q are WFFs, then $(P \wedge Q)$, $(P \vee Q)$, $(P \rightarrow Q)$ are WFFs.
4. If x is a variable and $P(x)$ is WFF, then $(\forall x P(x))$ and $(\exists x P(x))$ are also WFFs.

Example:

$(\forall x (P(x) \rightarrow Q(x))) \rightarrow \text{valid WFF.}$

📊 2.5 Inference in Predicate Logic

Inference means **deriving new knowledge** using predicates and quantifiers.

Example:

1. $(\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))) \rightarrow \text{All humans are mortal.}$
2. $(\text{Human}(\text{Socrates})) \rightarrow \text{Socrates is human.}$

$\therefore (\text{Mortal}(\text{Socrates})) \rightarrow \text{Socrates is mortal.}$

📌 3. Summary Table

Concept	Symbol / Meaning	Example
Proposition	Statement that is T/F	“ $2+2=4$ ”
Predicate	Statement with variable	$P(x): x > 5$
Universal Quantifier	$\forall x$ (for all x)	$\forall x (x^2 \geq 0)$

Existential Quantifier	$\exists x$ (there exists x)	$\exists x (x^2 = 4)$
Tautology	Always True	$P \vee \neg P$
Contradiction	Always False	$P \wedge \neg P$
Satisfiable	Sometimes True	$P \wedge Q$

4. Important Difference Tables

Concept	Propositional Logic	Predicate Logic
Basic Unit	Proposition	Predicate
Variables	None	Have variables
Quantifiers	Not used	Use $\forall, \exists$
Example	"It rains."	"x is even."
Scope	Simple true/false	Logical relationships

Concept	Tautology	Contradiction
Definition	Always true	Always false
Truth Table Output	All T	All F
Example	$P \vee \neg P$	$P \wedge \neg P$

Concept	Universal Quantifier ($\forall$)	Existential Quantifier ($\exists$)
Meaning	For all values	For at least one value
Truth Condition	True only if all cases true	True if any one case true
Example	$\forall x (x > 0)$	$\exists x (x > 0)$

Important Difference Tables – Unit III

1. Propositional Logic vs Predicate Logic

Basis	Propositional Logic	Predicate Logic
Basic Unit	Proposition (a simple statement)	Predicate (statement with variables)
Variables	Does not use variables	Uses variables (x, y, etc.)
Quantifiers	Not used	Uses $\forall$ (for all), $\exists$ (there exists)
Scope	Deals with complete truth statements	Deals with properties of objects
Example	"It is raining."	"x is greater than 5."
Power	Less expressive	More powerful, can express relationships

⚙️ 2. Tautology vs Contradiction vs Satisfiable

Basis	Tautology	Contradiction	Satisfiable
Definition	Always true for all inputs	Always false for all inputs	True for some values, false for others
Truth Table Output	All T	All F	Mixed (T and F)
Symbolic Example	$(P \vee \neg P)$	$(P \wedge \neg P)$	$(P \wedge Q)$
Circuit Meaning	Output always 1	Output always 0	Output depends on inputs

📊 3. Logical Connectives (AND vs OR vs NOT)

Basis	AND ($\wedge$)	OR ($\vee$)	NOT ($\neg$)
Operation Type	Conjunction	Disjunction	Negation
Output True When	Both inputs true	At least one input true	Input is false
Symbol	$\wedge$	$\vee$	$\neg$
Example	$(P \wedge Q)$	$(P \vee Q)$	$(\neg P)$
Truth Table Output (P=T, Q=F)	F	T	F

🔢 4. Universal Quantifier ($\forall$) vs Existential Quantifier ($\exists$)

Basis	Universal Quantifier ($\forall$)	Existential Quantifier ($\exists$)
Meaning	“For all”	“There exists at least one”
Truth Condition	True if all elements satisfy the condition	True if at least one element satisfies the condition
Symbol	$\forall$	$\exists$
Example	$\forall x (x^2 \geq 0) \rightarrow$ True for all x	$\exists x (x^2 = 4) \rightarrow$ True for x = 2, -2
Negation Rule	$\neg \forall x P(x) \equiv \exists x \neg P(x)$	$\neg \exists x P(x) \equiv \forall x \neg P(x)$

🗣️ 5. Statement vs Proposition

Basis	Statement	Proposition
Definition	Any sentence or expression	A statement that has definite truth value (T or F)
Truth Value	May or may not have	Must be either True or False
Example	“Close the door!” (not a proposition)	“The Earth is round.” (True)
Type	May be command/question	Always declarative

6. Simple Proposition vs Compound Proposition

Basis	Simple Proposition	Compound Proposition
Definition	A single statement with no connectives	Combination of two or more propositions
Operators Used	None	Uses AND, OR, NOT, $\rightarrow$, $\leftrightarrow$
Example	"5 < 7"	"5 < 7 and 7 < 9"
Complexity	Simple	Complex / combined statement

7. Logical Implication ($\rightarrow$) vs Biconditional ($\leftrightarrow$)

Basis	Implication ($\rightarrow$)	Biconditional ($\leftrightarrow$)
Meaning	"If P then Q"	"P if and only if Q"
Truth	False only when P = T and Q = F	True when both P and Q have same value
Expression	($P \rightarrow Q$)	($P \leftrightarrow Q$)
Example	If it rains $\rightarrow$ ground is wet	Ground wet $\leftrightarrow$ it rained

8. Algebra of Proposition vs Boolean Algebra

Basis	Algebra of Proposition	Boolean Algebra
Variables	Logical variables (T/F)	Binary variables (0/1)
Operators	$\wedge, \vee, \neg, \rightarrow, \leftrightarrow$	AND, OR, NOT
Application	Logic & reasoning	Circuit simplification
Output Type	Truth value (T/F)	Binary (1/0)
Example	($P \vee \neg P = T$)	($A + A' = 1$)

9. Inference vs Proposition

Basis	Proposition	Inference
Definition	Statement with a truth value	Logical conclusion from one or more premises
Truth	Known (T/F)	Derived by reasoning
Example	"It is raining."	"If it rains, ground gets wet $\rightarrow$ Ground is wet."
Used In	Logic statements	Logical reasoning / proofs

10. Well-Formed Formula (WFF) vs Not Well-Formed Formula

Basis	Well-Formed Formula (WFF)	Not Well-Formed Formula
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Definition	A syntactically correct logical formula	Expression that breaks logical syntax rules
Validity	Can be evaluated as True/False	Cannot be logically evaluated
Example	$(P \wedge Q) \rightarrow R$	$P \wedge \rightarrow Q$ (invalid form)
Structure	Correct use of connectives & brackets	Misplaced or missing symbols